

# NE523 HW 3

Kyle Hansen

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## Question 1 Distribution Functions

The flux energy spectrum is a distribution,

$$\psi(E) = 1/E, \quad (1)$$

while the cross sections are point functions,

$$\sigma_s(E) = \sigma_0, \quad (2)$$

$$\sigma_a(E) = \sigma_{a0} \sqrt{E_0/E}. \quad (3)$$

**1(a)** Since  $\phi$  is a distribution, it must be multiplied by  $|dE/du|$  to keep its integral properties under a change of variables. Lethargy is defined to be

$$u = \ln \left( \frac{E_0}{E} \right), \quad (4)$$

and

$$E = E_0 e^{-u}. \quad (5)$$

The derivative is

$$\frac{dE}{du} = -E_0 e^{-u} = -E. \quad (6)$$

Using these terms,

$$\phi(u) = \phi(E) \left| \frac{dE}{du} \right| \quad (7)$$

$$= \frac{1}{E} E \quad (8)$$

$$\phi(u) = 1. \quad (9)$$

The point functions can simply be transformed using the change of variables,

$$\sigma_s(u) = \sigma_0, \quad (10)$$

$$\sigma_a(u) = \sigma_{a0} \sqrt{E_0/E} \quad (11)$$

$$= \sigma_{a0} \sqrt{e^u} \quad (12)$$

$$= \sigma_{a0} e^{u/2}. \quad (13)$$

**1(b)** The function  $f_n$  is a distribution in  $E'$ , so must be transformed in a similar way to  $\phi$ ,

$$f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = f_n(E \rightarrow E', \hat{\Omega} \cdot \hat{\Omega}') \left| \frac{dE'}{du'} \right|. \quad (14)$$

Since  $\left| \frac{dE'}{du'} \right| = E'$ ,

$$f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \begin{cases} \frac{E'}{(1-\alpha)E} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}), & \alpha E \leq E' < E \\ 0, & \text{otherwise,} \end{cases} \quad (15)$$

where

$$\tilde{\mu} = \frac{1}{2} \left[ (A+1) \sqrt{\frac{E'}{E}} - (A-1) \sqrt{\frac{E}{E'}} \right] \quad (16)$$

and  $A$  is the atomic mass. Plugging in the  $E = E_0 e^{-u}$ ,

$$f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \begin{cases} \frac{E_0 e^{-u'}}{(1-\alpha)E_0 e^{-u}} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}), & \alpha E \leq E' < E \\ 0, & \text{otherwise,} \end{cases} \quad (17)$$

$$f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \begin{cases} \frac{e^{-(u'-u)}}{(1-\alpha)} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}), & \alpha E \leq E' < E \\ 0, & \text{otherwise,} \end{cases} \quad (18)$$

and transforming the boundaries,

$$f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \begin{cases} \frac{e^{-(u'-u)}}{(1-\alpha)} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}), & -u \ln \alpha \leq -u' < -u \\ 0, & \text{otherwise,} \end{cases} \quad (19)$$

or

$$f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \begin{cases} \frac{e^{-(u'-u)}}{(1-\alpha)} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}), & u < u' < u \ln \alpha \\ 0, & \text{otherwise.} \end{cases} \quad (20)$$

**1(c)** Since  $f_n$  is only nonzero within the bounds  $u < u' < u \ln \alpha$ , the integral bounds can be changed,

$$\int_{-\infty}^{\infty} du' \int_{4\pi} d\hat{\Omega}' f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \int_u^{u \ln \alpha} du' \int_{4\pi} d\hat{\Omega}' f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}'). \quad (21)$$

So using Equation 20,

$$\int_{-\infty}^{\infty} du' \int_{4\pi} d\hat{\Omega}' f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \int_u^{u \ln \alpha} du' \int_{4\pi} d\hat{\Omega}' \frac{e^{-(u'-u)}}{(1-\alpha)} \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}) \quad (22)$$

$$= \frac{e^u}{1-\alpha} \int_u^{u \ln \alpha} du' e^{-u'} \int_{4\pi} d\hat{\Omega}' \delta(\hat{\Omega} \cdot \hat{\Omega}' - \tilde{\mu}). \quad (23)$$

From the definition of the dirac delta function, the integral over angle equals one, so

$$\int_{-\infty}^{\infty} du' \int_{4\pi} d\hat{\Omega}' f_n(u \rightarrow u', \hat{\Omega} \cdot \hat{\Omega}') = \frac{e^u}{1-\alpha} \int_u^{u \ln \alpha} du' e^{-u'} \quad (24)$$

$$= \frac{e^u}{1-\alpha} \left[ -e^{-u'} \right]_{u'=u}^{u \ln \alpha} \quad (25)$$

$$= \frac{e^u}{1-\alpha} \left[ -\alpha e^u + e^u \right] \quad (26)$$

$$= \frac{2}{1-\alpha} \left[ -\alpha + 1 \right] \quad (27)$$

$$= 1. \quad (28)$$

## Question 2 Uncollided Flux

In a homogenous, source-free slab of thickness  $a$ , the transport equation for uncollided flux only (no scattering source) is

$$\mu \frac{d}{dx} \psi_0(x, \mu, E) + \Sigma(E) \psi_0(x, \mu, E) = 0, \quad 0 < x < a, \quad (29)$$

with vacuum boundary given by

$$\psi_0(a, \mu, E) = 0, \quad \mu < 0, \quad (30)$$

and the incoming flux boundary condition given by

$$\psi_0(0, \mu, E) = I \delta(\mu - \tilde{\mu}) \delta(E - E_0), \quad \mu > 0. \quad (31)$$

The ODE is solved with separation of variables,

$$\mu \frac{d\psi_0}{dx} = -\Sigma_t(E) \psi_0 \quad (32)$$

$$\int \frac{1}{\psi_0} d\psi_0 = - \int \frac{\Sigma_t(E)}{\mu} dx \quad (33)$$

$$\ln \psi_0 = \frac{-\Sigma_t x}{\mu} + c_0 \quad (34)$$

$$\psi_0(x, \mu, E) = c_1(\mu, E) e^{-\frac{\Sigma_t}{\mu} x}. \quad (35)$$

The incoming flux boundary conditions are used to find  $c_1(\mu, E)$ ,

$$\psi_0(0, \mu, E) = c_1 = I \delta(\mu - \tilde{\mu}) \delta(E - E_0), \quad \mu > 0, \quad (36)$$

$$\psi_0(a, \mu, E) = c_1 e^{-\frac{\Sigma_t}{\mu} a} = 0 \quad (37)$$

$$c_1(\mu, E) = 0, \quad \mu < 0. \quad (38)$$

Using these values of  $c_1$ , the solution for the uncollided flux is thus

$$\psi_0(x, \mu, E) = I \delta(\mu - \tilde{\mu}) \delta(E - E_0) e^{-\frac{\Sigma_t}{\mu} x}. \quad (39)$$

## Question 3 Once-collided flux

Applying the scattering operator to Equation 39 yields

$$q_{s,1} = \int_0^\infty dE' \int_{-1}^1 d\mu' \Sigma_s(E' \rightarrow E, \mu_0) \psi_0(x, \mu', E'), \quad (40)$$

where  $\mu_0$  is the scattering angle. To avoid using  $\mu_0$ , this can also be expanded using the legendre moments of the scattering cross section,

$$q_{s,1} = \sum_{l=0}^{\infty} (2l+1) P_l(\mu) \int_0^\infty dE' \Sigma_{s,l}(E' \rightarrow E) \phi_{l,0}(x, E') \quad (41)$$

where  $\Sigma_{s,l}$  is the cross section projected onto the  $l$ th Legendre polynomial, and  $\phi_{l,0}$  is the  $l$ th Legendre moment of  $\psi_0$ ,

$$\phi_{l,0} = \int_{-1}^1 d\mu' \frac{1}{2} P_l(\mu') \psi_0(x, \mu', E, t) \quad (42)$$

$$= \frac{1}{2} I P_l(\tilde{\mu}) \delta(E - E_0) e^{-\frac{\Sigma_t}{\tilde{\mu}} x}. \quad (43)$$

Plugging back into the equation for the source,

$$q_{s,1} = \frac{1}{2} I \sum_{l=0}^{\infty} (2l+1) P_l(\mu) P_l(\tilde{\mu}) \int_0^\infty dE' \Sigma_{s,l}(E' \rightarrow E) \delta(E - E_0) e^{-\frac{\Sigma_t}{\tilde{\mu}} x} \quad (44)$$

$$= \frac{1}{2} I \sum_{l=0}^{\infty} (2l+1) P_l(\mu) P_l(\tilde{\mu}) \Sigma_{s,l}(E_0 \rightarrow E) e^{-\frac{\Sigma_t(E_0)}{\tilde{\mu}} x} \quad (45)$$

and the equation for once-collided flux is

$$\mu \frac{d}{dx} \psi_1(x, \mu, E) + \Sigma_t(E) \psi_1(x, \mu, E) = \frac{1}{2} I \sum_{l=0}^{\infty} (2l+1) P_l(\mu) P_l(\tilde{\mu}) \Sigma_{s,l}(E_0 \rightarrow E) e^{-\frac{\Sigma_t(E_0)}{\tilde{\mu}} x} \quad (46)$$

$$\psi_1(0, \mu, E) = 0, \quad \mu > 0 \quad (47)$$

$$\psi_1(a, \mu, E) = 0, \quad \mu < 0, \quad (48)$$

where both boundary conditions are vacuum, since the beam contributes only to uncollided flux.

## Question 4 Point source

The integral transport equation for isotropic scattering and isotropic source is

$$\phi(\mathbf{r}, E) = \int_V dV' \frac{\exp[-\tau(E, \mathbf{r}' \rightarrow \mathbf{r})]}{4\pi |\mathbf{r}' - \mathbf{r}|^2} \left[ \int_0^\infty dE' (\Sigma_s(\mathbf{r}', E' \rightarrow E) + \chi(E)\nu(E')\Sigma_f(\mathbf{r}', E')) \phi(\mathbf{r}', E') + S(\mathbf{r}', E) \right], \quad (49)$$

and in the case of zero scattering and zero fission, there is no secondary source and the equation becomes

$$\phi(\mathbf{r}, E) = \int_V dV' \frac{\exp[-\tau(E, \mathbf{r}' \rightarrow \mathbf{r})]}{4\pi |\mathbf{r}' - \mathbf{r}|^2} S(\mathbf{r}', E). \quad (50)$$

In a homogenous medium, the optical length is

$$\tau(E, \mathbf{r} - \rho' \hat{\Omega} \rightarrow \mathbf{r}) = \int_0^{\rho'} d\rho'' \Sigma_t(\mathbf{r} - \hat{\Omega} \rho'', E) \quad (51)$$

$$= \Sigma_t(E) \rho'. \quad (52)$$

Then let  $r' = |\mathbf{r}' - \mathbf{r}|$ . Using this optical length, and plugging in the point source, the integral transport equation becomes

$$\phi(\mathbf{r}, E) = \int_V dV' \frac{\exp[-\Sigma_t(E)r']}{4\pi r'^2} S(E) \delta(\mathbf{r}'). \quad (53)$$

Integrating the dirac delta yields

$$\phi(\mathbf{r}, E) = \frac{\exp[-\Sigma_t(E)r]}{4\pi r^2} S(E), \quad (54)$$

where  $r = |\mathbf{r}|$ .